



Team Name:

***MEMORA***

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UNIVERSITI SAINS MALAYSIA

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Title of business idea:

**MEMORA**

“A bracelet for your family that  
monitoring elderly health's and provide sweet memory legacy.”

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## 1.0 INTRODUCTION

In today's rapidly aging world, a silent epidemic is emerging—elderly individuals passing away alone, unnoticed for days or even weeks. While this tragedy is global in scope, countries like Singapore highlight the urgency: in 2023, at least 37 seniors were found deceased in their homes, with no one around to help or even notice their passing. By 2030, over 9.2% of Singapore's elderly population is expected to live alone, mirroring trends seen in Japan, the U.S., and across Europe.

This is not just a social issue—it's a deeply emotional one. As families grow smaller and urban lifestyles create physical distance, many children worry silently about the well-being of their aging parents or grandparents. Yet, current solutions in the market are often either too simplistic or too clinical, failing to address both the practical safety needs and the emotional gap that modern families face.

Enter Memora—a next-generation safety bracelet designed to monitor, protect, and preserve. Equipped with advanced health sensors, movement detection, geofencing, and emotional well-being features, Memora is more than a device—it is a symbol of love. Even in absence, it allows families to stay connected, reassured, and close to the ones they hold dear.

**Our mission is to deliver global market an intelligent, emotionally aware safety wearables that protect the elderly, ease caregiver worries, and preserve the bonds between generations.** Through this mission, we aim to drive meaningful change—not just in how we care, but in how we remember, connect, and love.

**Our vision is simple: no elderly person should be left unseen, unheard, or forgotten.** Memora offers both technological peace of mind and a lasting emotional legacy—making it a product with heart, purpose, and global relevance.

## 2.0 PROBLEM STATEMENT

According to Department of Statistic of Malaysia (DOSM), Malaysia is rapidly moving toward becoming an aging nation, with the elderly (aged 60 and above) expected to make up over 15% of the population by 2030. However, this shift is not

matched with adequate infrastructure, social support, or technology designed to serve this growing demographic. Another study by DOSM also estimates indicate that between **7.4% and 10%** of Malaysians aged 60 and above live alone. Another report from Sinar Daily also reported that in Perlis, a state with a population under 300,000, there were **80 recorded cases** of elderly individuals dying alone between 2022 and August 2024.

Many elderly Malaysians, particularly those in the B40 income group or suffering from dementia, face problem such as **Social isolation**, especially those living alone. **Increased risk of health emergencies**, including unnoticed accidents and deteriorating conditions. **Limited access to real-time assistance**, due to mobility issues, distance from family, or lack of communication tools. **Financial constraints**, affecting access to quality care and lifestyle improvements. These issues raise urgent questions about how to preserve **dignity, autonomy, and safety** for aging individuals.

In line with the theme “**Empowering Elderly Lives,**” we aim to develop **MEMORA** – a next-generation smart safety bracelet that have the following feature. **Monitors health vitals** (heart rate, fall detection, etc.). **Tracks real-time location** to prevent elderly wandering or getting lost. **Provides emergency alerts** to caregivers and family members instantly. **Offers long battery life and is dementia-patient friendly** (tamper-proof and sturdy)

Our goal is to improve the **overall well-being** of the elderly, reduce emergency response time, and restore peace of mind to both the elderly and their families. This solution supports the vision of a future where elderly individuals can age **gracefully and independently**.

### 3.0 OVERALL BUSINESS CONCEPT

Memora bracelet is a revolutionary wearable technology designed to redefine care, connection, and peace of mind for elderly individuals, people with cognitive impairments, and their caregivers. Positioned at the intersection of cutting-edge health technology and emotional wellness, **Memora bracelet offers two unique value propositions: first, comprehensive health and safety monitoring; and second, emotional and memory-based connectivity.** These core pillars not only address practical caregiving challenges but also enhance emotional bonds within families, preserving both well-being and legacy.

From a healthcare perspective, this Memora bracelet stands out as a multifunctional, real-time monitoring tool equipped with an array of advanced sensors. These include a **body temperature sensor, blood oxygen monitor, and motion detection system.** The **Fall Detection Sensor, which uses an accelerometer and gyroscope combination,** ensures immediate alerts in the event of sudden falls, which is critical for elderly safety. Furthermore, the **Sleep Quality Sensor analyzes various sleep stages which is light, deep, and REM**—to help caregivers understand recovery patterns, adjust routines, and ensure the individual is truly resting well. These features make the bracelet a 24/7 health companion, reducing risk and improving proactive caregiving.

In terms of location and mobility, **Memora bracelet features a GPS system** comparable to that of Apple's iWatch, offering more precise, real-time tracking. This is especially valuable for families with dementia patients or young children. The **emergency SOS button** subtly embedded in Memora will trigger when Memora detected a sudden critical vital change or if Memorian leaves the safe zone for too long or too far, Memorian exact location will be recorded and our system will instantly make a phone call to caregiver, if caregiver fail to answer the call, our system will immediately contact with cooperated hospital. Additionally, **Geofencing Technology** enables caregivers to define “safe zones” like home, or care centers, and automatically sends alerts if the user exits these zones. These features are crucial for preventing wandering or getting lost, often a significant concern in aging or neurodivergent populations.

Memora bracelet's emotional value is equally compelling. Through the **Family/Caregiver Sync** feature, all health data is connected to an intuitive mobile app that allows multiple users—children, grandchildren, or hired caregivers—to monitor vital signs and movements remotely. The bracelet also offers a **Medication Reminder System** using vibration and light notifications, helping users take medications on time without confusion. But what truly sets this device apart is its **Digital Legacy Diary**—a sentimental feature that allows users or their families to upload voice notes, personal messages, or memories to the app, all of which are synced with the bracelet's usage history. During memorable festival such as mother and father days, Memora will surprise caregiver will memorable memory of Memorian. This creates a living archive of love and remembrance. To ensure our client data privacy, we created our own server with end-to-end encryption with all the data secure in a cloud storage. To align with sustainability, we avoid the use of one time battery and make use of rechargeable Li-Po battery to be more eco-friendly.

The business model of the Memora bracelet relies on hardware sale and tiered subscription service system that supports accessibility and personalization. For hardware, we offer two product which are Memora Gen 1 for basic feature and Memora Gen 1 Pro for advanced feature. We provide monthly subscription: Silver, Gold, and Diamond plan. Higher-tier plans offer additional caregiver linkages, more memory storage, monthly wellness reports, “Legacy Letters,” memory capsules, and point multipliers. These plans drive continuous revenue.

In summary, this Memora bracelet is not just a wearable device—it's a bridge between health, safety, and human connection. It **combines medical-grade monitoring, innovative safety features, and deeply personal emotional tools** in one sleek, easy-to-use product. With a subscription-based model that accommodates various needs and budgets, and a promise of legacy through digital memory keeping, this product stands to revolutionize how we care for our aging population—both physically and emotionally.

## 4.0 SITUATION ANALYSIS

### 4.1 SWOT ANALYSIS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>STRENGTHS</u></b><br><br>1. <b>Continuous Healthcare Monitoring:</b> Collect information concerning vital signs like heart rates, blood pressure and falls.<br>2. <b>Added Safety:</b> Real time detection of falls, alerts for emergencies allows the families, caregivers to live with peace of mind.<br>3. <b>Memory Inheritance:</b> Enables seniors to record, share life stories.                                                                                                            | <b><u>WEAKNESS</u></b><br><br>1. <b>Smartphone Dependency:</b> Full functionality in smartphone may be hindered for elders. Limitation to administering these devices in the elder's category.<br>2. <b>Battery Life Issues:</b> The day-to-day charging requirements might be considered a burden for older individuals.<br>3. <b>Cost Problems:</b> Too costly for older persons with fixed incomes.                                                                                                                |
| <b><u>OPPORTUNITIES</u></b><br><br>1. <b>Improved Conditions at elderly homes:</b> Automation of health tracking and emergency alerts considerably lightens the burden on the shoulder of caregivers.<br>2. <b>Support in Dementia and Alzheimer's:</b> Security and considerably decreased worry among patients with cognitive impairment brought visibility in tracking the location.<br>3. <b>Better Communication with Families:</b> Provides real time updates on the health status of a loved one. | <b><u>THREATS</u></b><br><br>1. <b>Competition from Other Markets:</b> The massive influx of health wearable devices on the market itself appears to threaten the differentiation of a single device and its securing of market share.<br>2. <b>Cultural Constraints:</b> Some elders may be apprehensive about using new technology-based device.<br>3. <b>Economic Recessions:</b> During economic recessions, the first thing that gets hit is discretionary purchases like health devices, which means low sales. |

## 4.2 COMPETITOR ANALYSIS

1. Empatica EmbracePlus: An FDA- and CE-approved medical-grade smartwatch that offers 24/7 monitoring of vital signs—heart rate, skin temperature, and motion—with real-time data sharing via the Empatica Care platform, ideal for seniors at risk of seizures or respiratory issues.
2. CardiacSense CSF-3 Watch: A CE-marked and FDA-class-approved medical watch that continuously monitors heart-related metrics like heart rate, oxygen in blood, and ECG. It is non-invasive in type, designed for elderly persons, which is perfect for those having cardiovascular diseases. Reports can be given monthly and for data-sharing purposes with medical physicians.
3. Movano Evie Ring: FDA approved, this smart ring is enticingly affordable at \$269 (almost RM1,185.62) with no subscription fees yet simple with tracking heart rate, skin temperature and sleep patterns. Yet, its battery discharges less than 3 days, while the app runs on iOS 16 or higher, making it not so widely accessible to all older adults.

## 4.3 COMPETITIVE ADVANTAGE

1. **Continuous Monitoring & Comprehensive Health-Tracing:** Being recorded and accounted for activity, pace, localization, heart rate, blood oxygen levels, temperature, rest, and sleep, which provide a picture of the health status of the individual and their daily activity.
2. **Data Capture & Storage:** The health data gets continuously captured and stored securely so that it can be accessed and monitored for long periods.
3. **Analytics Dashboard:** It brings in place an intuitive dashboard to users, which gives real-time data to monitor their health trends all the time.
4. **Reporting Functionality:** Provides users, and their clinicians, with all such reports that will assist in guiding smart decisions and personalized care plans.
5. **Heritage of Memory:** Enables users to inscribe shared memories, messages, or unique occasions with the idea of making digital heritage that can be transferred to generations after generations to encourage emotional wellness and relations between people.

## 5.0 MARKETING STRATEGY



5.1 PRODUCT STRATEGY

Memora bracelet is more than a wearable—it's an all-in-one **health guardian**, **safety alert system**, and **digital keepsake** for families with aging loved ones. What sets Memora bracelet apart in a saturated market:

| Key Features |                                                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.           | <b>Advanced Health Sensors:</b> Oxygen, blood pressure, heart rate variability (HRV), and UV sensors.                                              |
| 2.           | <b>Safety First:</b> Real-time GPS tracking, fall detection (accelerometer + gyroscope), and emergency SOS button.                                 |
| 3.           | <b>Daily Wellness:</b> Sleep staging, movement tracking, and medication reminders.                                                                 |
| 4.           | <b>Damage-resistant Build:</b> Designed to withstand accidental impacts, ideal for dementia or Parkinson’s patients.                               |
| 5.           | <b>Emotional Legacy:</b> <i>Digital Legacy Diary</i> —a platform for families to preserve voice notes, memories, and moments linked to the device. |

This fusion of **emotional connection** and **functional innovation** creates a product that people don't just buy—they bond with.

5.2 PRICE STRATEGY

We adopt a **value-based pricing** model targeted at the **middle- and upper-class market** segments, offering:

| Memora Product       |                                                                                 | Subscription   |                                                                                                                      |
|----------------------|---------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------|
| Memora Gen 1: RM 699 | Essential monitoring features + max1 person monitoring                          | Silver RM 40   | Max caregiver linkage:2 + 256GB with memory diary storage + wellness history report in a years +x1.5point multiplier |
| Gen 1 pro: RM 1199   | Max 5 person monitoring + 256GB memory storage + max 3 generation memory legacy | Gold RM 60     | Max caregiver linkage:4 + 512GB with wellness history report in 3 years + x2 point multiplier                        |
|                      |                                                                                 | Diamond RM 100 | Max caregiver linkage:8 + 1TB + wellness history report in 5 years + x3 point multiplier + Legacy letter Capsule     |

5.3 DISTRIBUTION STRATEGY

To reach our target audience efficiently and credibly, our distribution plan includes:

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| Online D2C Channels  | Official website and major e-commerce platforms (Shopee, Lazada, Amazon). |
| Healthcare Retailers | Partnerships with pharmacies and eldercare centers.                       |
| Medical Institutions | Pilot programs with hospitals, rehab clinics, and dementia homes.         |
| CSR Collaborations   | Partner with NGOs and eldercare NGOs for subsidized units and awareness.  |

This hybrid approach ensures we meet customers where they are—both physically and emotionally.

5.4 PROMOTION STRATEGY

|                         |                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------|
| Emotive Video Marketing | Short films on real-life cases of elderly isolation.                                 |
| Influencer Outreach     | Collaborate with doctors, eldercare advocates, and family influencers.               |
| Digital Advertising     | Targeted Facebook & Instagram ads using behavioral data (age, caregiving interests). |
| Referral Program        | Families who refer others receive discounts or lifetime cloud backup.                |
| Public Health Tie-ins   | Co-campaigns during World Alzheimer’s Month or National Seniors’ Day.                |

By blending **rational reassurance** with **emotional storytelling**, we aim to create a movement—not just a market.

## 6.0 IMPLEMENTATION AND OPERATIONAL PLAN

### 6.1 Overview of Phases & Timeline Structure

| Phase                     | Duration           | Dates (2025)  | Lead        |
|---------------------------|--------------------|---------------|-------------|
| 1. Preparation & Design   | 2 months           | May–Jun       | Product     |
| 2. Hardware Finalization  | 3 months           | Jul–Sep       | Engineering |
| 3. Software Development   | 4 months (overlap) | Jun–Nov       | Software    |
| 4. QA & Pilot Launch      | 2 months           | Dec–Jan ('26) | QA & Ops    |
| 5. Full Rollout & Scaling | 3 months           | Feb–Apr ('26) | Marketing   |
| 6. Final Evaluation       | 1 month            | May ('26)     | Executive   |

*Phase 3 overlaps Phases 1–2 to accelerate launch; final evaluation happens after 6 full months of data.*

### 6.2 OEM & No-Store Model

- **OEM Partnership**
  - **Selection (May):** Vet and contract with 2 to 3 Tier-1 wearable-device OEMs with Li-Po battery and sensor integration capabilities.
  - **Design Handoff (Jun):** Deliver PCB schematics, mechanical enclosure CAD, sensor spec sheet.
- **No Physical Store**
  - **E-Commerce Setup (May–Jun):** Build Shopify-based storefront; integrate payment gateway and subscription management.
  - **Fulfillment (Jul onward):** Partner with 3PL warehouse in KL for pick-pack and direct-to-customer shipping.

### 6.3 Software Programming Roadmap

|   | Dates         | Deliverable                            | Responsible   |
|---|---------------|----------------------------------------|---------------|
| 1 | Jun 1–Jul 15  | Mobile app wireframes; API spec        | UX/Backend    |
| 2 | Jul 16–Aug 31 | Core SDK: sensor data ingestion + sync | Firmware/S DK |
| 3 | Sep 1–Oct 15  | Geofencing + SOS alert module          | Backend       |

|   |               |                                          |          |
|---|---------------|------------------------------------------|----------|
| 4 | Oct 16–Nov 30 | Memory Diary UI + cloud storage security | Frontend |
|---|---------------|------------------------------------------|----------|

• **Tech Stack:**

- Mobile: React Native
- Backend: Node.js + AWS (Lambda, DynamoDB, S3)
- SDK/Firmware: C++/FreeRTOS
- Security: AES-256 end-to-end encryption

## 6.4 Gantt Chart and Milestones

| Task                          | Progress (%) | Responsible Team           | Resources                                   | Resources needed                                             | Timeline (month) |   |   |   |   |   |   |   |   |         |  |  |
|-------------------------------|--------------|----------------------------|---------------------------------------------|--------------------------------------------------------------|------------------|---|---|---|---|---|---|---|---|---------|--|--|
|                               |              |                            |                                             |                                                              | 1                | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | ongoing |  |  |
| 1. <b>Product Development</b> | 90           | Project Lead               | 1.Lee Teck Qing<br>2.Lim Tian Yang          | Project Manager, Dev tools, PM Software                      |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Hardware Development        | 100          | Hardware Engineering       | 1.Liew Jian Sheng<br>2.Johnson Lim Ther How | Sensors, PCBs, IoT tools, testing lab                        |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Software Development        | 85           | Software & App Engineering | 1.Yap Tian Sheng<br>2.Liew Jian Sheng       | IDEs, cloud platform, mobile app frameworks (e.g., Flutter)  |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Integration of Features     | 80           | Cross-functional Team      | 1.Lee Teck Qing<br>2.Johnson Lim Ther How   | API bridge tools, testing hardware, firmware modules         |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Prototype Testing           | 70           | QA / Testing               | 1.Lim Tian Yang                             | Test cases, QA tools, test users, bug tracking system        |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Regulatory & Compliance     | 50           | Compliance & Legal         | 1.Liew Jian Sheng                           | Legal consultants, compliance checklist, data security tools |                  |   |   |   |   |   |   |   |   |         |  |  |
| 2. <b>Pilot Program</b>       | 30           | Operations & Field Team    | 1.Lee Teck Qing                             | Devices, demo kits, monitoring tools, user training material |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Select Pilot Sites          | 100          | Business Development       | 1.Yap Tian Sheng                            | Contact lists, CRM system, stakeholder presentations         |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Deploy Bracelets            | 50           | Field Operations           | 2.Johnson Lim Ther How                      | Shipping tools, local setup crew, onboarding documentation   |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Monitor Performance         | 20           | Analytics & Support        | 1.Liew Jian Sheng                           | Analytics dashboard, user feedback tools, support agents     |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Iterate and Update          | 10           | Product Development Team   | 1.Lee Teck Qing<br>2.Johnson Lim Ther How   | Developer time, bug fixes, updated firmware/software         |                  |   |   |   |   |   |   |   |   |         |  |  |
| 3. <b>Full Rollout</b>        | 0            | Executive & Sales          | 1.Yap Tian Sheng                            | Budget, logistics partners, marketing assets                 |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Mass Production             | 0            | Manufacturing Team         | 1.Yap Tian Sheng<br>2.Johnson Lim Ther How  | Contract manufacturer, component inventory, QA station       |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Sales & Distribution        | 0            | Sales & Marketing          | 1.Liew Jian Sheng<br>2.Lim Tian Yang        | Sales team, marketing materials, CRM tools                   |                  |   |   |   |   |   |   |   |   |         |  |  |
| - Training & Support          | 0            | Customer Success           | 1.Lee Teck Qing<br>2.Lim Tian Yang          | Support portal, training videos, live onboarding sessions    |                  |   |   |   |   |   |   |   |   |         |  |  |

## Key Milestones:

1. **June 30** – OEM tooling approved & e-commerce live
2. **August 31** – SDK v1.0 (data ingestion)
3. **October 15** – SOS/geofence tested in staging
4. **November 30** – Full app + memory diary ready
5. **December 15** – Pilot cohort (n=50) onboarded
6. **February 15 ('26)** – 1,000 units shipped; 60% subs rate
7. **May 1 ('26)** – KPI review & strategic next steps

## 6.5 Workload Distribution & Task Allocation

| Task                                         | Team / Role          |
|----------------------------------------------|----------------------|
| <b>OEM negotiations &amp; procurement</b>    | Product Manager      |
| <b>Mechanical &amp; PCB design revisions</b> | Hardware Engineering |
| <b>Firmware integration &amp; testing</b>    | Firmware Engineers   |

|                                                   |                        |
|---------------------------------------------------|------------------------|
| <b>API design &amp; backend services</b>          | Backend Engineers      |
| <b>Mobile app UI/UX</b>                           | UX Designer + Frontend |
| <b>Cloud infra &amp; security compliance</b>      | DevOps / Security      |
| <b>E-comm store build &amp; subscription flow</b> | Ops / E-Commerce Lead  |
| <b>QA test plans &amp; pilot support</b>          | QA Engineers / Ops     |
| <b>Marketing launch campaign</b>                  | Marketing Manager      |
| <b>Logistics &amp; 3PL coordination</b>           | Supply-Chain Lead      |
| <b>Customer support setup</b>                     | Customer Success Team  |

Each phase has a “**Phase Lead**” who coordinates sub-teams, reports weekly, and escalates blockers.

## 6.6 Feasibility & Executive Oversight

### 1. Budget Checkpoints:

- End of Phase 2: hardware tooling + software dev burn rate
- End of Phase 4: cost per pilot user vs. projected LTV

### 2. Risk Mitigation:

- Dual-sourcing critical components
- Rolling-release feature flags in software

### 3. Executive Reviews (monthly):

- **Steering Committee** chaired by CEO, includes heads of Product, Engineering, Finance.
- Approve go/no-go at Milestones 2, 4, and 6.

## 4. Data Management

| Area                | Description                                                               |
|---------------------|---------------------------------------------------------------------------|
| Data Collection     | Continuous streaming from device to cloud or local server                 |
| Data Storage        | Secure, encrypted storage following GDPR/industry standards               |
| Analytics Dashboard | Role-based dashboards for employees, supervisors, and HR                  |
| Reporting           | Automated reports on safety compliance, activity trends, and fatigue risk |

## 5. Maintenance & Support

| Task             | Description                                               |
|------------------|-----------------------------------------------------------|
| Firmware Updates | Over-the-air updates for performance and security patches |
| Customer Support | 24/7 helpdesk and ticketing system for issue resolution   |

|                      |                                                            |
|----------------------|------------------------------------------------------------|
| Hardware Replacement | RMA system for device faults or wear-and-tear replacements |
|----------------------|------------------------------------------------------------|

## 6.7 Final Evaluation

**May 2026:**

- **KPIs:**
  - Units shipped vs. target
  - Subscription conversion & churn
  - Mean time to alert & incident resolution
  - User NPS & memory-diary engagement
- **Executive Report:**
  - Detailed P&L vs. forecast
  - Engineering retrospectives (velocity, bugs)
  - Go-forward recommendations (e.g., ASEAN expansion, feature roadmap)

## 7.0 FINANCIAL PLANNING

### 7.1 Start Up Capital

We will begin with 1,200 units Memora for S, M, L size to reduce initial risk and test market response. A start-up capital of RM347,200.00 will cover manufacturing through OEM, app development, branding, and operations.

#### Memora Bracelet – Manufacturing Cost per Unit (in MYR)

| No     | Category           | Component                         | Unit Cost (MYR) |
|--------|--------------------|-----------------------------------|-----------------|
| 1      | Microcontroller    | ESP32-WROOM-32                    | RM18.80         |
| 2      | Heart Rate Sensor  | MAX30102                          | RM11.75         |
| 3      | Fall Detection     | MPU6050                           | RM7.05          |
| 4      | GPS Module         | NEO-6M                            | RM23.50         |
| 5      | Thermal Sensor     | MLX90614                          | RM32.90         |
| 6      | Battery            | LP402025                          | RM7.05          |
| 7      | Protection IC      | DW01 + FS8205                     | RM1.88          |
| 8      | Voltage Regulator  | AMS1117-3.3                       | RM0.94          |
| 9      | N-Channel MOSFET   | IRLZ44N / AO3400                  | RM1.41          |
| 10     | P-Channel MOSFET   | IRF9540N / SI2301                 | RM1.41          |
| 11     | PCB                | Custom PCB                        | RM14.10         |
| 12     | Passive Components | Capacitors, resistors, etc.       | RM2.35          |
| 13     | Bracelet Coating   | Polycarbonate + TPU + UV printing | RM10.00 (est.)  |
| 14     | Shock Absorber     | Silicone gel pad                  | RM5.00 (est.)   |
| 15     | Assembly & Testing | Labour + QC                       | RM5.00 (est.)   |
| 16     | Packaging          | Box + User Manual + Design        | RM5.00 (est.)   |
| 17     | Size Customization |                                   | RM3.00 (est.)   |
| 18     | Assembly & Testing |                                   | RM5.00 (est.)   |
| TOTAL: |                    |                                   | Rm155.94        |

Total Estimated Manufacturing Cost per Unit for Basic unit (Memora Gen 1) is RM155.94 ~ RM156.00. For advanced unit (Memora Gen 1 Pro) we estimate allocate RM300.00 more for advanced features. Cost per unit for Memora Gen 1 Pro is RM186.00. We produce 600 units of Memora Gen 1 and 600 units Memora Gen 1 Pro.

#### Start up Capital Summary

| Category                        | Description                                                                   | Cost (MYR) |
|---------------------------------|-------------------------------------------------------------------------------|------------|
| Manufacturing Cost              | Memora Gen 1 (RM156.00 x 600 units)<br>Memora Gen 1 Pro (RM186.00x 600 units) | 205,200.00 |
| App Development (iOS + Android) | Cross-platform app with memory diary, alerts                                  | 45,000.00  |
| Backend Development             | API, geofencing logic, health data encryption                                 | 30,000.00  |
| Cloud Server Setup              | Setup & configuration (Memora private server)                                 | 15,000.00  |
| 1-Year Hosting & Maintenance    | Bandwidth, security updates, SSL, DB backups                                  | 10,000.00  |

|                           |                                                     |            |
|---------------------------|-----------------------------------------------------|------------|
| Business Registration     | Legal, SSM, IP/trademark filing                     | 5,000.00   |
| Marketing Materials       | Branding, launch videos, posters                    | 8,000.00   |
| Launch Event & PR         | Venue, booth design, demo kits                      | 7,000.00   |
| Team & Admin Budget       | Part-time developer/assistant (3 months)            | 10,000.00  |
| Market Research / Testing | Surveys, interviews, usability test kits            | 2,000.00   |
| Miscellaneous & Buffer    | Unexpected costs, warranty returns, app maintenance | 10,000.00  |
| TOTAL                     |                                                     | 347,200.00 |

## Funding

We seek RM350,000.00 in start-up capital to launch Memora, covering production, app development, staffing, branding, and legal setup. Funding will come from personal contributions, angel investors, and grants.

| Funding Source                    | Amount (RM)    | Percentage  | Details                                                                   |
|-----------------------------------|----------------|-------------|---------------------------------------------------------------------------|
| Personal Investment               | 52,500         | 15%         | Founder's contribution to demonstrate commitment and reduce investor risk |
| Angel Investor(s)                 | 245,000        | 70%         | In exchange for equity or convertible notes                               |
| Government Grants / Startup Funds | 52,500         | 15%         | Apply to technology or elderly-focused innovation grants                  |
| <b>Total Capital Raised</b>       | <b>350,000</b> | <b>100%</b> |                                                                           |

## 7.2 Budget

In Year 1, we'll scale production efficiently to support growth and brand awareness. After the initial start-up of total 1200 units in first quarter, we plan to produce total of 3600 more units: 1,200 units for following quarter of year in Q2, Q3, and Q4 (200units for every single size in a quater). The budget will be expanded in following year as production of product increases, aligned with rising demand, market expansion, and festive opportunities.

| No. | Category                             | Description                                                                       | Estimated Cost (MYR) |
|-----|--------------------------------------|-----------------------------------------------------------------------------------|----------------------|
| 1   | <b>Manufacturing Cost</b>            | Memora Gen 1 (RM156.00 x 1,800 units)<br>Memora Gen 1 Pro (RM186.00x 1,800 units) | 615,600.00           |
| 2   | App and Server Maintenance & Updates | Server hosting, bug fixing, new features, security patches                        | 18,000.00            |
| 3   | Customer Support & Admin             | 1-2 part-time staff for tech/customer service, admin costs                        | 24,000.00            |



|       |                                  |                                                                |            |
|-------|----------------------------------|----------------------------------------------------------------|------------|
| 4     | Marketing & Outreach             | Online campaigns, influencer collabs, awareness programs       | 30,000.00  |
| 5     | Events & CSR Campaigns           | Family day events, elderly awareness, memory capsule campaigns | 10,000.00  |
| 6     | Logistics & Packaging            | Shipping materials, custom boxes, manuals                      | 15,000.00  |
| 7     | Subscription Platform Management | Payment gateway, storage scaling, billing system support       | 6,000.00   |
| 8     | Business Development             | Travel, partnerships (eg. hospitals, NGOs), pilot testing      | 10,000.00  |
| 9     | Reserve/Contingency Fund         | Unexpected overhead, repairs, price shocks                     | 10,000.00  |
| TOTAL |                                  |                                                                | 738,600.00 |

### 7.3 Revenue (Short Term)

As of 2024, Department of Statistic Malaysia (DOSM) states that Penang has approximately **143,805 residents aged 65 and above**, accounting for about **8% of the state's population**. There is a demand for Memora, we estimate that around 10% of Penang will own and aware with Memora bracelet. We projected all products are completely sold in first year. We projected that 50% of our customer will purchase for monthly subscription plan. Total revenue for first year is RM4,651,200.

| Sales                    | Units | Unit Price (RM) | Total Revenue (RM) |
|--------------------------|-------|-----------------|--------------------|
| Memora Gen 1             | 2,400 | 699             | 1,677,600          |
| Memora Gen 1 Pro         | 2,400 | 1,199           | 2,877,600          |
| Silver Subscription Plan | 2400  | 40              | 96,000             |
| <b>Total Sales</b>       | —     | —               | <b>4,651,200</b>   |

### 7.4 Revenue (Long Term)

As of 2024, Malaysia has approximately **3.9 million people aged 60 and above**, representing **11.6%** of the total population of 34.1 million. As in Penang, Data from the Department of Statistics Malaysia (DOSM) has revealed that Penang will have the largest proportion of people aged 60 and above in Malaysia at 26.2% by 2040, an increase from 14.87% in 2020. This show that Memora have potential market demand in the future. In terms of elderly care, there are **393 registered elderly care centres and 26 nursing homes in 2024 by UNDP**. In the future of 3-5 years,

Memora aims to collaboration with nursing homes, hospitals, insurance providers, and government agencies to supply bracelets in bulk for eldercare. These partnerships allow for bulk sales (estimate 50–200 units per deal). Memora aim to caregiver dashboards for monitoring multiple users.

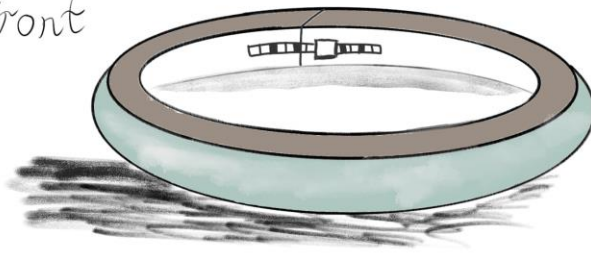
| Year | Product Sales<br>(Units Sold)                                                         | Subscription Revenue<br>(Estimated Users)     | B2B/Institutional<br>Collaboration Revenue                       | Total Estimated<br>Revenue (RM) |
|------|---------------------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------|---------------------------------|
| 2025 | 4,800 units →<br><b>RM4,555,200</b><br>(2,400 Gen 1 @ RM699<br>+ 2,400 Pro @ RM1,199) | 50% of users (2,400) →<br><b>~RM96,000</b>    | Not applicable                                                   | <b>RM4,651,200</b>              |
| 2026 | 5,500 units →<br><b>~RM5,225,000</b>                                                  | 65% of users (3,575) →<br><b>~RM1,300,000</b> | Total 50 facilities @ avg<br>RM10,000/year →<br><b>RM500,000</b> | <b>RM7,025,000</b>              |
| 2027 | 6,500 units →<br><b>~RM6,325,000</b>                                                  | 70% of users (4,550) →<br><b>~RM1,600,000</b> | Total 100 facilities →<br><b>RM1,000,000</b>                     | <b>RM8,925,000</b>              |
| 2028 | 7,500 units →<br><b>~RM7,275,000</b>                                                  | 75% of users (5,625) →<br><b>~RM2,250,000</b> | Total 150 facilities →<br><b>RM1,500,000</b>                     | <b>RM11,025,000</b>             |
| 2029 | 9,000 units →<br><b>~RM8,550,000</b>                                                  | 80% of users (7,200) →<br><b>~RM3,000,000</b> | Total 200 facilities →<br><b>RM2,000,000</b>                     | <b>RM13,550,000</b>             |

## 8.0 PROTOTYPE

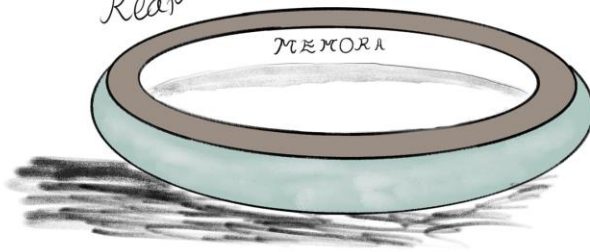
### 8.1 Memora Design Model

Retro

Front

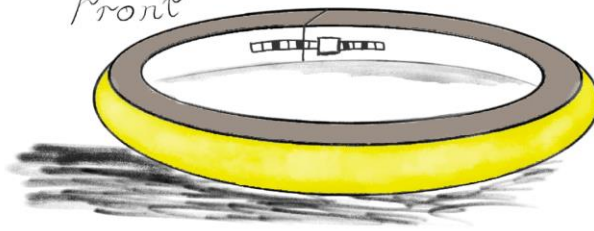


Rear

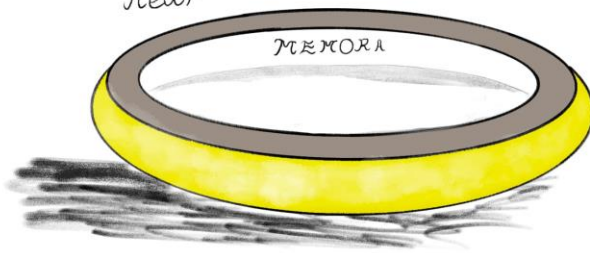


Luxury

Front

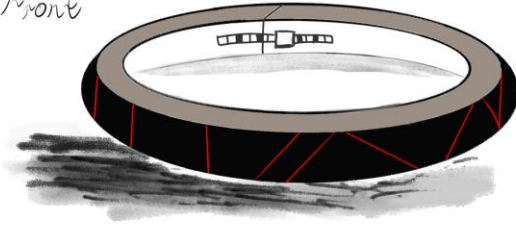


Rear

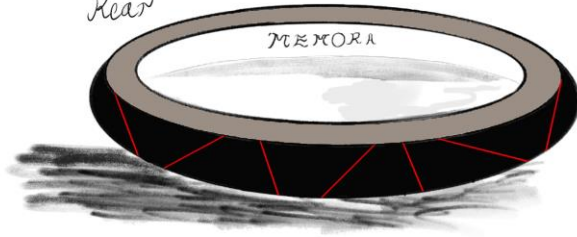


*Sport*

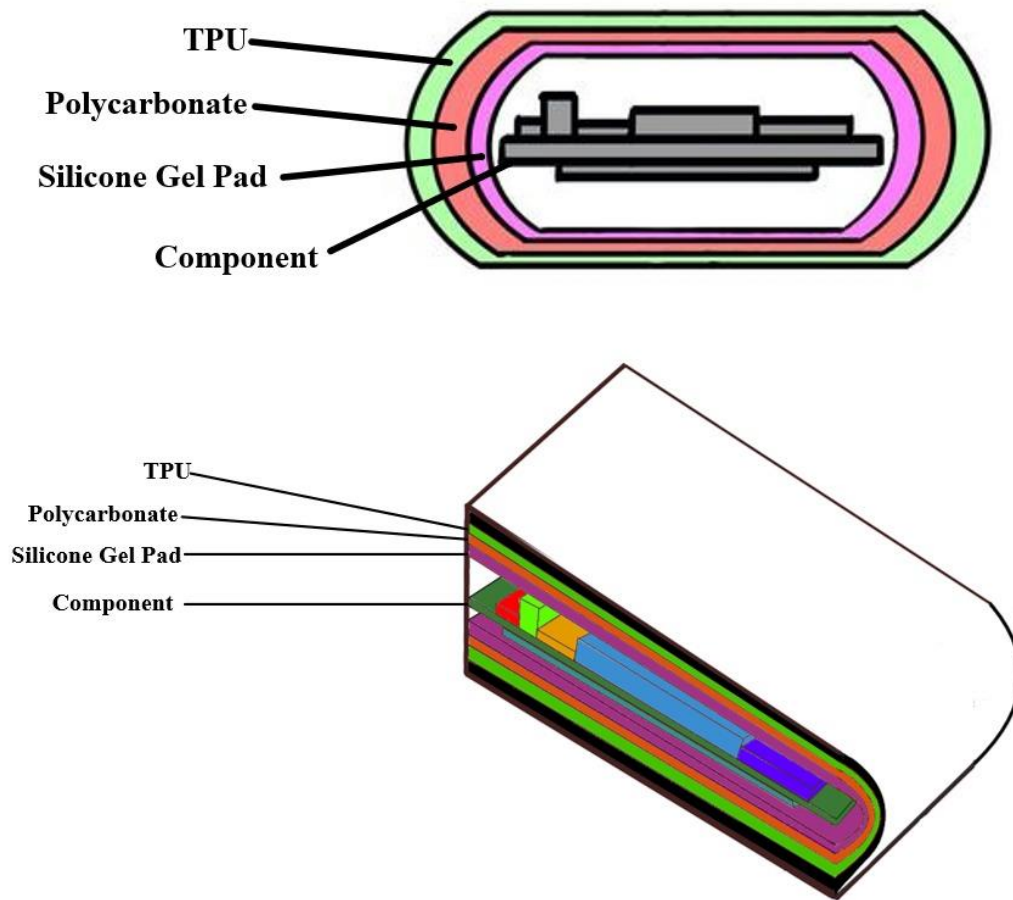
*Front*



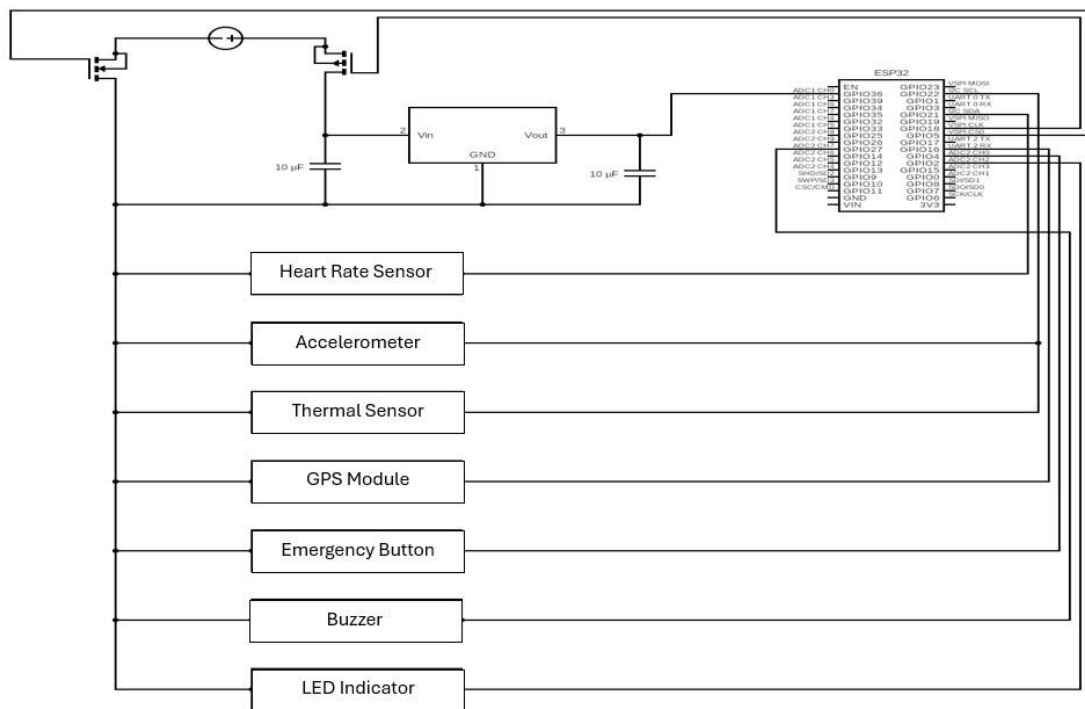
*Rear*



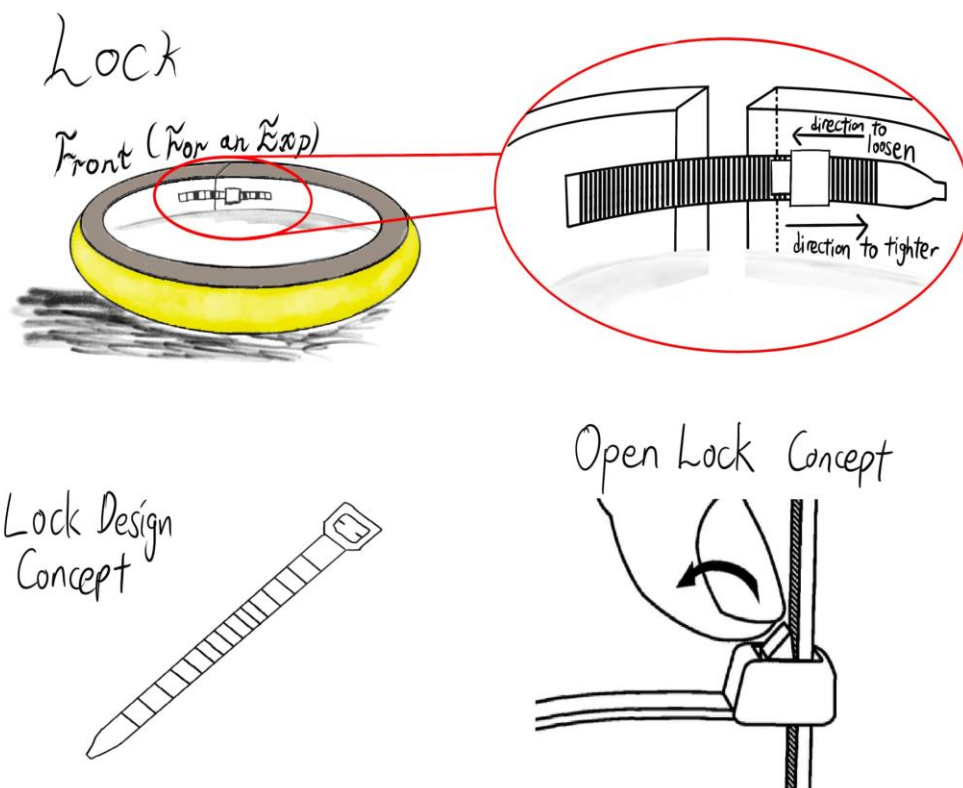
## 8.2 Cross Sectional Drawing



### 8.3 Schmetic Diagram

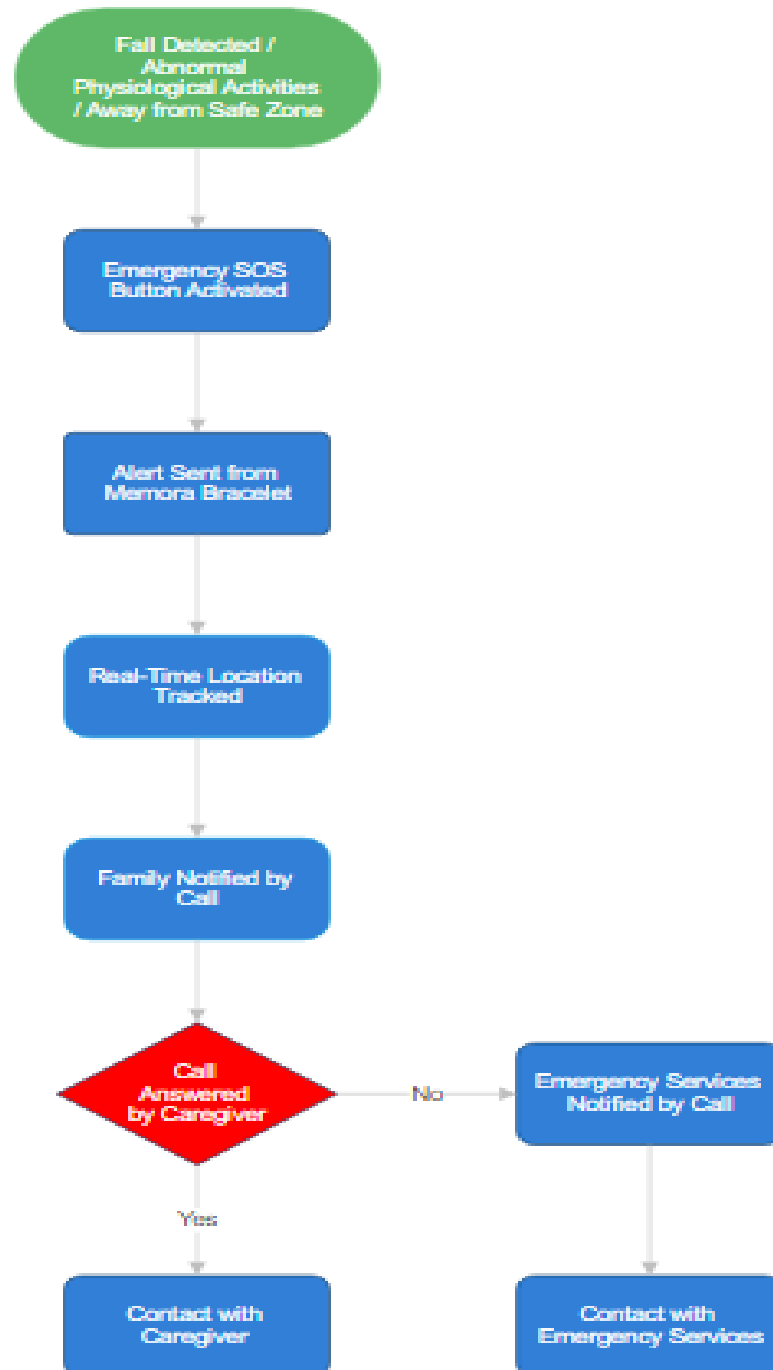


### 8.4 Lock System

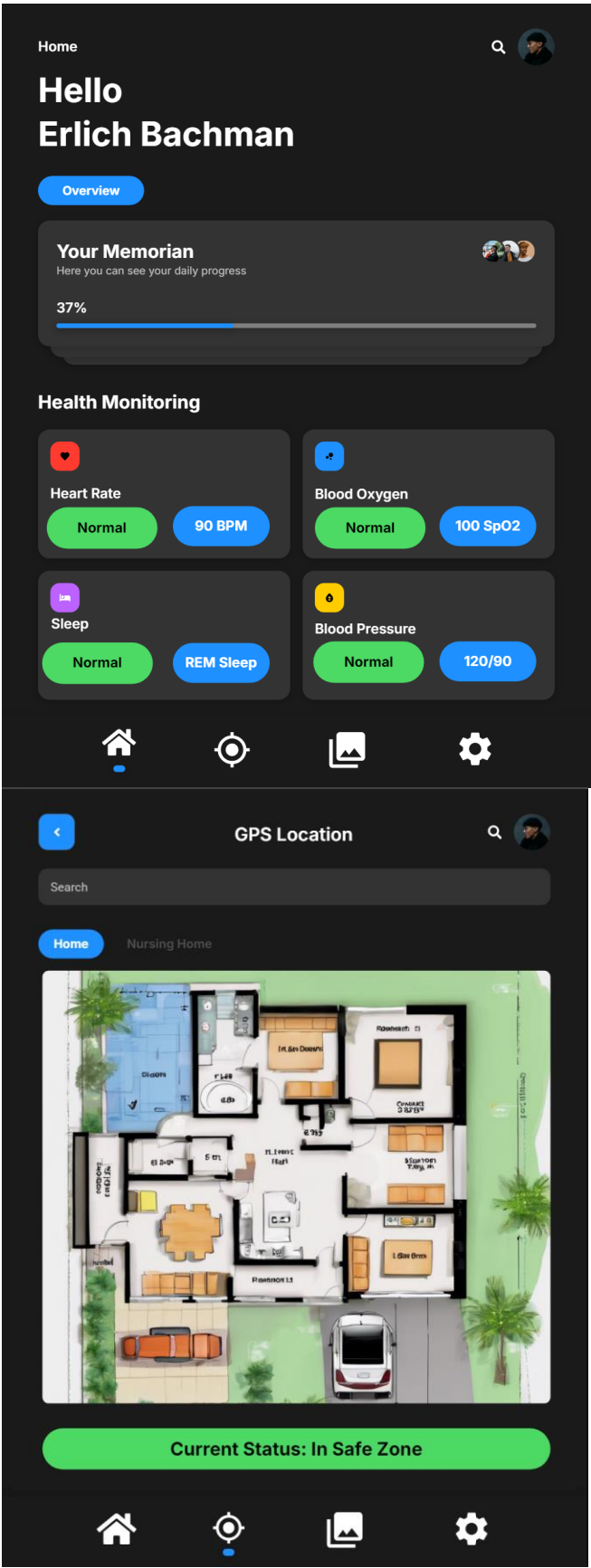


## 8.5 Emergency SOS Flow Diagram

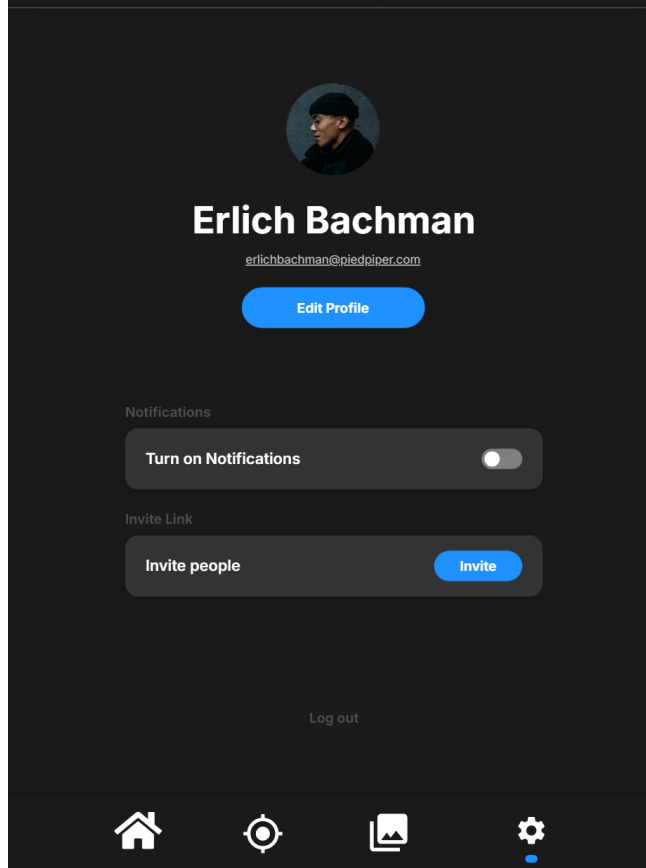
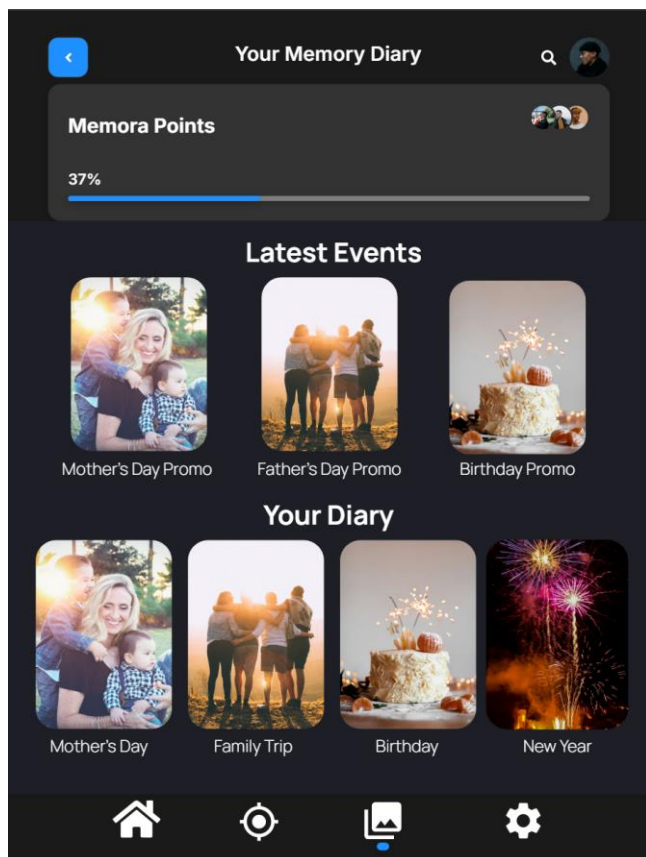
# Emergency SOS Flow Diagram



8.6 APP Design







APPENDIX

Draft Design

Market Research



Malaysia, Singapore & Brunei  
Darussalam

WHO WE AREWHAT WE DOOUR IMPACTGET INVOLVED



Expanding Care Infrastructure with Community Integration

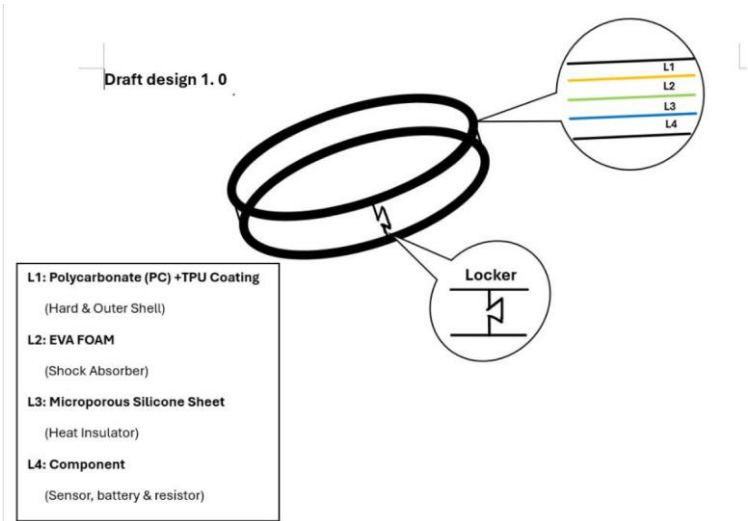
There are currently 393 registered elderly care centres and 26 nursing homes (7), with estimates indicating between 700 to over 1000 unregistered facilities (8). Even with most families preferring to provide caregiving for the elderly at home, many facilities already face long waiting lists, and the demand will only increase in the coming years. Initiatives under the 12th Malaysia Plan aimed at fostering senior-friendly communities through activity centres (Pusat Aktiviti Warga Emas) and healthcare facilities are steps in the right direction (9). Recognising the severe shortage of elderly care facilities, Malaysia must also accelerate the development of care options.

GEORGE TOWN, Sept 7 (Bernama) -- The Penang government has launched the Age-Friendly City: Penang Island Strategic Action Plan, setting the course for the state to become age-friendly by 2028.

State Local Government, Town and Country Planning Committee chairman H'ng Mooi Lye said the initiative involves a tripartite collaboration between the Penang Island City Council (MBPP), the Penang Women's Development Corporation (PWDC) and the Malaysian Healthy Ageing Society (MHAS) consultants.

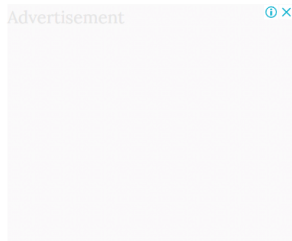
He said the initiative, which began in 2022, consists of eight strategic modules outlined by the World Health Organisation (WHO) to meet the criteria for Age-Friendly City status.

"Based on the Department of Statistics Malaysia's 2024 projections, 8 per cent or 143,805 of Penang's population will be senior citizens (aged 65 and above), with 73,975 living on the island and 69,830 in Seberang Perai.



A recent Sinar report revealed a growing number of such cases.

In Perlis alone, a state with fewer than 300,000 residents, 80 solitary deaths were recorded between 2022 and August 2024.



According to Persatuan Sains Sosial Malaysia (PSSM) special advisor Professor Emeritus Datuk Dr Abdul Rahman Embong, United Nations (UN) data from 2021 ranked Malaysia as the third-highest Asean country in terms of unmarried individuals, at 44.7 per cent.

He warned that Malaysia's rapidly ageing population is on track to becoming an aged nation before achieving developed status, leading to an increasing number of people living alone.



TheStar

Elderly man dies at home, alone

**NATION**

Monday, 10 Jul 2023

**Related News**



**NATION** 07 Apr 2025

**'Better to keep watch  
than regret later if  
something goes missing'**

MUAR: He was found lying motionless on a bed in the living room of his house where he lived alone.

It was the third day of Hari Raya Aidilfitri, an occasion which he should have been celebrating.

Instead, his neighbours detected an odour coming from his home. The 75-year-old man was believed to have been dead for at least three days before his decomposed body was found at about 11pm on April 24.

It was an example of how some elderly folk fall through the cracks, unnoticed by others.

The Star reported last week that there were almost 10 cases in the past few months of senior citizens found dead at home where they lived alone.

According to the National Health and Morbidity Survey 2018, an estimated 7.4% of elderly people lived alone.

